Eduardo **Rodríguez Sánchez**

Computer Engineer · HPC Master Student

Madrid, Spain

contact@erodriguez.dev | https://erodriguez.dev/ | nekronos-gh | eduardo-rodriguez-sanchez

Education

Univ. of Luxembourg / Sofia Univ. St. Kliment Ohridski

Belval / Sofia

M.S. in High Performance Computing — EUMaster4HPC double degree

Sep. 2025 - Present

- Focus: parallel algorithms, HPC system architecture, accelerator programming, and performance analysis.
  GPA: 17.5/20.

UC3M / Georgia Institute of Technology

Madrid / Georgia, USA

B.S. in Computer Engineering (exchange year: Georgia Tech)

Sep. 2020 - Jun. 2024

- Coursework: Operating Systems, Networks, Distributed Systems, Machine Learning, Cryptography.
  GPA: 9.1/10 (UC3M) · 4.0/4.0 (Georgia Tech).
  Thesis: C++ load-balancing proxy sustaining 86 Gb/s of trace forwarding across millions of concurrent connections.

Skills

Programming	C, C++, Python
HPC / Systems	OpenMP, MPI, AVX2, Multithreading, Profiling, Linux, GDB, Valgrind
Distributed / Cloud	Kubernetes, Spark, 5G Networking, gRPC, Prometheus, Grafana
Languages	Spanish (native), English (fluent)

Experience

Ericsson

Madrid, Spain

Software Developer

Dec. 2024 - Aug. 2025

- Shipped User Plane Analytics features and antenna geo-redundancy for the 5G Core, and cut multi-threaded lock contention to resolve critical production issues.
- Led the migration of the 5G RIB codebase to the Madrid site and redesigned intra-forwarding-module communication to run entirely inside Kubernetes, eliminating external network hops.

Samsung Electronics (Zhilabs)

Madrid, Spain

Software Engineer

Jun. 2023 - Oct. 2024

- Built a network trace proxy sustaining 86 Gb/s with a recursive-descent binary parser for trace analysis, and parallelized a multi-antenna traffic simulator to speed up testing.
- Designed a RAN Assistant, a proof-of-concept multi-agent LLM/NLP/ML system for telecom Q&A, anomaly detection, and automated root-cause analysis.

Research

SnT (Security, Reliability and Trust)

Luxembourg

Research Assistant

Oct. 2025 - Apr. 2026

- Prototyped a Byzantine fault-tolerant PBFT consensus engine in C++ on an async networking stack, including a cross-replica-group coordination protocol.
- Built failure-injection scenarios and reproducible benchmarking tooling to validate consensus performance under adversarial conditions.

Projects

Rule110 Fast

github.com/nekronos-gh/rule110

Cellular-automaton simulator vectorized with AVX2 and parallelized with OpenMP, using aligned memory in modern C++23.

TinyFile

github.com/nekronos-gh/TinyFile

Zero-copy file-compression service built on POSIX message queues and shared memory, driven by a ring-buffer state machine.

PBFT

github.com/nekronos-gh/pbft-cpp

Full 3-phase consensus implementation in C++ (Salticidae) with view changes and checkpointing, instrumented via Prometheus/Grafana.